

Target-Specific Recoverability Maps for Reduced-Lead ECG Reconstruction

A graded per-lead identifiability + conditioning certificate with calibrated intervals, and an honest account of a retracted fabrication claim

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Abstract

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers from one truncated SVD of the segment’s empirical rank-3 spatial sub-matrix—a graded *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$. The per-lead criterion is the classical estimability condition specialized to ECG; the contribution is its use as a graded, *empirically-estimated* map (the estimated subspace differs 43–55° from the textbook inverse-Dower dipole), the distribution-free calibrated intervals we attach to it, and a graded ST-safety study. On PTB-XL the map ranks target leads on a continuum by normalized identifiability and expected mV ambiguity—from limb leads, anterior precordial ST (V1–V4) is far from identifiable while lateral V5/V6 are much less ambiguous (none exactly identifiable)—and the total ST-threshold error rate is empirically similar across the evaluated reconstructors (45.8–47.3%; an empirical observation, not a certified bound) while the false-positive/false-negative split is a reconstructor choice. The certificate transfers across PTB-XL and Chapman for QRS (aligned subspaces, reproduced κ ordering) but not in the ST/T third direction. Finally, we document in full a claim this project retracted—that a diffusion reconstructor proves “fabrication is a property of the objective”—which a leakage-corrected, multi-seed re-analysis reduced to a marginal ($\approx 1.3\sigma$) effect with no supporting realism trend. This long version accompanies a four-page conference paper and contains the full proofs, ablations, cross-dataset study, and the negative result.

1 Introduction

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads, and a reconstruction that gets the mean right but invents an ST deviation invents a finding.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead’s low-rank morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis $\mathbf{M}_s$ (top-3 singular vectors of the population lead covariance). We stress that $\mathbf{M}_s$ is a *population-estimated (PCA)* object: on real PTB-XL it shares two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 5).

This document is the extended version of a four-page conference submission. It adds: the full statement and proof of the certificate (Sec. 3, App. A); the cross-dataset transfer study (Sec. 6); and a complete account of a claim we retracted during pre-submission review (Sec. 7). We report the retraction in detail on purpose: hiding the earlier, more exciting numbers would be precisely the selective reporting our certificate is designed to expose.

2 Related Work

Lead reconstruction and VCG synthesis. Fixed linear transforms synthesize the vectorcardiogram or missing leads from a standard set (inverse-Dower [3], Kors regression [4]), and reduced-lead reconstruction has a long clinical literature [5]; recent deep and arbitrary-mask waveform-completion models push reconstruction accuracy [1, 2, 6, 7]. These target the reconstruction; we target the *prior* question of what a given configuration can support, independent of the reconstructor. Accordingly our neural baseline is a *representative* masked-completion U-Net of this family, included to show the certificate binds even a flexible learned reconstructor—not a claim to beat [6, 7]. **Identifiability and lead selection.** Whether a linear functional is recoverable from a sub-observation is the classical estimability question [8]; choosing which leads to observe is optimal sensor selection [9]. Our per-lead η is estimability specialized to the ECG dipolar functional; the novelty is not the identity but the graded, empirically-estimated, per-target-lead map with calibrated per-record error attached. **Uncertainty for inverse problems.** Conformal prediction [10, 11] gives distribution-free intervals under exchangeability; we apply Mondrian CQR per (S, s, ℓ) group.

3 The Recoverability Certificate

Model. Within segment s write the population 12-lead vector as $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$, where $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, $\mathbf{d}$ the dipole coordinates, and $\mathbf{r}_s$ the non-dipolar residual. Observing S gives $\mathbf{y}_S = \mathbf{Sel}_S \mathbf{L}_s + \mathbf{n}$. Let $\mathbf{M}_{s,S} = \mathbf{Sel}_S \mathbf{M}_s$ have truncated SVD at relative tolerance ϱ ; $\mathbf{M}_{s,S}^+$ its truncated pseudo-inverse and $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$ the projector onto the observed dipole coordinate directions.

Proposition 1 (Per-target-lead dipolar identifiability and conditioning). *The mean-centred dipolar estimate $\hat{\mathbf{L}}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{y}_S - \boldsymbol{\mu}_{s,S})$ satisfies, for each target lead ℓ with $\mathbf{e}_\ell$ the ℓ -th unit vector:*

- (i) *Identifiability. Define $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S : $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ is determined by $\mathbf{Sel}_S \mathbf{M}_s \mathbf{d}$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ ; its dipolar component is not identifiable at any SNR.*
- (ii) *Conditioning. On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$: $|\mathbf{e}_\ell^\top (\hat{\mathbf{L}}_s - \boldsymbol{\mu}_s - \mathbf{M}_s \mathbf{P}_{\text{obs}} \mathbf{d})| \leq \kappa_{s,\ell}(S) (\|\mathbf{Sel}_S \mathbf{r}_s\| + \|\mathbf{n}\|)$.*

The global $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ is a configuration-level (spectral) worst case: since a row norm never exceeds the spectral norm, $\kappa_{s,\ell}(S) \leq \kappa_s(S)$ for every ℓ , so κ_s upper-bounds every per-lead conditioning (it is not their maximum). Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend on the estimated $\mathbf{M}_s$ and on ϱ ; near-rank-deficient S are reported with an ϱ -sensitivity sweep and record-bootstrap confidence intervals.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [8]) specialized to the per-lead linear functional $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$: $\eta_{s,\ell}(S) = 0$ iff that functional lies in the row space of $\mathbf{M}_{s,S}$, and $\kappa_{s,\ell}$ is its coefficient norm. We claim no novelty for this identity. The contribution (Sec. 5) is its use as a *graded*, per-target-lead, *empirically-estimated* recoverability map: a continuous η that ranks target leads (not a binary rank test), a subspace $\mathbf{M}_s$ that differs materially from the textbook inverse-Dower dipole, and per-record calibrated error rates (Sec. 4) attached to it—quantities not read off lead geometry.

The non-dipolar residual (no independence claim on real data). Let $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s \mid \mathbf{y}_S]$ and $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$, so $\mathbb{E}[\mathbf{u}_s \mid \mathbf{y}_S] = \mathbf{0}$. We do *not* assume $\mathbf{u}_s \perp \mathbf{y}_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|^2 = \mathbb{E}[\text{Var}(\mathbf{r}_s \mid \mathbf{y}_S)].$$

Part of $\mathbf{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\mathbf{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—*not* declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only). *If the data are generated so that a component $\mathbf{u}$ is statistically independent of $\mathbf{y}_S$ (an explicit synthetic construction), then $p(\mathbf{u} \mid \mathbf{y}_S) = p(\mathbf{u})$, the Bayes estimate of $\mathbf{u}$ is its prior mean, and $\mathbb{E}\|\hat{\mathbf{u}} - \mathbf{u}\|^2 \geq \text{Var}(\mathbf{u})$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

4 Distribution-Free Calibration

4.1 Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment’s samples). We train a gradient-boosted quantile regressor of it on the observed leads’ segment means and apply Mondrian conformalized quantile regression [10] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor’s `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha=0.90$). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI [0.89, 0.93], mean width 0.150 mV).

What is and is not guaranteed. Mondrian CQR gives *within- (S, s, ℓ) -group marginal* coverage under exchangeability—*not* coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ($n=85$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

Table 1: Limb-6 $\rightarrow$ anterior-precordial ST reconstruction, 1498 fold-10 records. FP = false-positive, FN = false-negative ST-threshold crossings (%); total = FP+FN. The total is empirically similar across reconstructors; the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	14.5	32.8	47.3	0.076
ridge	6.9	38.9	45.8	0.072
OLS	6.2	39.6	45.9	0.073

5 Experiments

Data. PTB-XL [12] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [13]; the map is a deterministic function of the estimated $\mu_s, \mathbf{M}_s$ (record-level bootstrap CIs), not a held-out evaluation. All table numbers regenerate from result JSON via `paper/emit_baseline_table.py`.

5.1 The recoverability map on PTB-XL

We estimate $\mathbf{M}_s$ (rank 3) per segment on the training NORM records and report the map with RECORD-LEVEL bootstrap CIs (each resample draws whole records, re-pools their segment samples, and refits $\mathbf{M}_s$; truncation tolerance $\varrho=10^{-2}$; Fig. 1). The map is a deterministic function of the estimated $\mathbf{M}_s$ and the selection S ; the CI quantifies its estimation uncertainty. A single observed lead leaves every other lead’s dipolar component unidentifiable; a dipole-spanning triplet {I,II,V2} or the five-lead {I,II,V1,V3,V5} makes all targets *exactly* identifiable ($\eta=0$ to machine precision, since S spans the rank-3 subspace).

Exact vs. graded identifiability. Exact identifiability is the binary test $\eta_{s,\ell}(S)=0$. For limb-6, *no* precordial lead is exactly identifiable ($\eta_{ST,\ell}>0$ for all), so we grade the shortfall with two scale-aware quantities rather than raw η : the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell} / \|\mathbf{e}_\ell^\top \mathbf{M}_s\|_2 \in [0, 1]$ (the fraction of the lead’s dipolar content that is unobservable) and the *expected unobserved ambiguity* in mV under the fitted dipole-coordinate prior. Both fall from strongly unrecoverable anterior leads (V2 $\tilde{\eta}=0.960$, ambiguity 0.207 mV; V3, V1, V4) to less-ambiguous lateral leads (V5 $\tilde{\eta}=0.170$, 0.025 mV; V6 $\tilde{\eta}=0.057$, 0.008 mV). We therefore say **anterior precordial ST (V1–V4) is far from identifiable from limb leads, while lateral V5/V6 are much less ambiguous—but none is exactly identifiable**. A physiological precordial/limb split cannot make this graded call; the empirical map does.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads kept exact; ST measured at J+60 ms vs. a PR baseline on fiducials located once on the observed Lead II and shared between truth and reconstruction), and count ST-threshold events ($|\text{ST}| \geq 0.1$ mV, elevation or depression; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). The *total* wrong-event rate is *empirically similar* across the evaluated reconstructors (45.8–47.3%); all three *miss* more true crossings than they *invent* (FN>FP)—limb $\rightarrow$ precordial reconstruction smooths ST toward the population and drops real deviations—and the FP/FN balance shifts with the reconstructor (the dipolar recovery has the largest false-positive share, OLS the smallest). We report the whole matrix; because we do not derive a minimax/Bayes lower bound, we describe the shared total as an empirically similar total error rate across these three reconstructors—*not* a certified floor.

Truncation-tolerance sensitivity. Rank/ κ of well-posed configurations are stable across

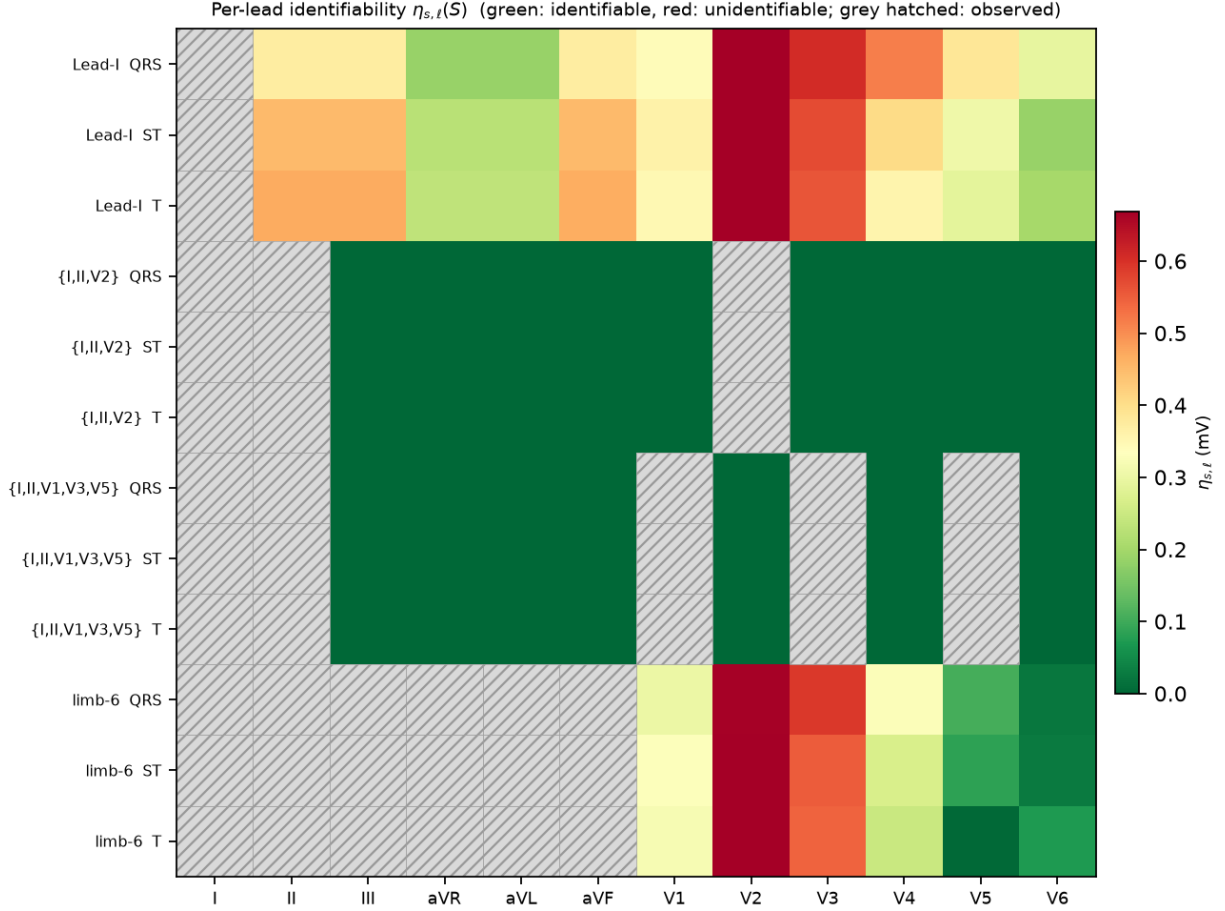


Figure 1: Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). For limb-6 the anterior precordial leads are the most unidentifiable; lateral V5/V6 are less ambiguous.

$\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ (e.g. $\{I, II, V1, V3, V5\}$ is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data $\{V1, V2, V3\}$ is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa \sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ -sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

5.2 Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge $\rightarrow$ U-Net step is within its paired CI)—prior-mean $\rightarrow$ dipolar (Tier I) $\rightarrow$ per-segment ridge (Tier I+II)

Table 2: Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.467	0.236	0.220	0.183
{I,II,V1,V3,V5}	0.481	0.183	0.167	0.164
limb-6	0.481	0.361	0.344	0.238

→ U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on {I,II,V1,V3,V5}; paired Δ CI $[-0.007, +0.000]$ mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481 → 0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46\times$ its spanning-set error (paired spanning→limb-6 Δ CI $[+0.070, +0.079]$ mV) and no reconstructor closes that gap. This residual gap is *consistent with* the unobserved dipolar coordinate the map’s $\eta > 0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket “not recoverable.”

Empirical, not physical, subspace. On real PTB-XL the estimated $\mathbf{M}_s$ shares two directions with the classical inverse-Dower vectorcardiographic space [3, 4] (principal angles $2\text{--}6^\circ$) but its third direction differs materially (max angle $43\text{--}55^\circ$, bootstrap CIs excluding small angles). We therefore treat $\mathbf{M}_s$ as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

6 Cross-Dataset Transfer of the Certificate

The certificate is a function of the estimated subspace $\mathbf{M}_s$ and the selection S ; a fair question is whether it is an artifact of one cohort. We re-estimate $\mathbf{M}_s$ and the per-configuration conditioning on the independent Chapman-Shaoxing 12-lead database [14] (different country, devices, and sampling) and compare to PTB-XL (`results/cross_dataset.json`); the re-estimation uses 349 Chapman records ($\sim 13.8\text{k}$ segments per segment type). We deliberately do *not* claim “population independence”; we report what actually transfers.

QRS subspace is transfer-stable. For QRS the two datasets’ rank-3 subspaces are nearly aligned (principal angles $20.3^\circ/9.1^\circ/1.6^\circ$; explained-variance ratios 0.56/0.27/0.08 vs. 0.55/0.27/0.06). The per-configuration conditioning transfers for the well-posed sets: κ for {I,II,V2} is 3.1 (PTB-XL) vs. 4.1 (Chapman) and for {I,II,V1,V3,V5} 2.5 vs. 2.3, and the limb-6 ill-conditioning is reproduced on both ($\kappa \sim 2 \times 10^5$). The near-rank-deficient {V1,V2,V3} transfers less cleanly (3.9 vs. 6.7, a $\sim 70\%$ increase), as expected for an ill-conditioned set. So the qualitative identifiable/unidentifiable verdict and the κ ordering are stable for well-posed configurations, with larger cohort variation on the near-degenerate ones.

ST/T third direction does not transfer. For the low-amplitude segments (P, ST, T) two of three subspace directions align but the third differs sharply between datasets (max principal angle 88–89°; P 88.4°, ST 88.3°, T 89.2°). We therefore restrict the transfer claim: the QRS map and the limb-6 conditioning verdict are cohort-stable; the ST/T *third* spatial direction is cohort-specific and its map should be re-estimated per dataset. This is the honest scope of transfer, not a blanket invariance.

Sensitivity to duplicated limb leads (results/lead_weighting.json). The six limb leads are exact combinations of I and II, so a 12-lead PCA up-weights frontal-plane content. Refitting $\mathbf{M}_s$ on the 8 algebraically-independent leads and lifting back via the fixed transform $\mathbf{T}$, the *identifiability verdict is robust*: for the spanning sets the per-lead η is unchanged (max $|\Delta\eta|=0$; all targets remain exactly identifiable), and limb-6 precordial leads stay unidentifiable under both fits. The *magnitudes* of the low-amplitude ST/T subspace are, however, sensitive: the 12-lead and 8-lead ST/T subspaces differ by large principal angles (75°/82°) and κ can shift several-fold (e.g. ST {I,II,V2}: 3.8 → 38). We therefore read the QRS map quantitatively and the ST/T *graded* values qualitatively—the binary identifiable/unidentifiable verdict is robust; the exact ST/T η/κ /ambiguity magnitudes are weighting-dependent.

7 A Negative Result: the Generative “Fabrication-Objective” Story Did Not Survive

This project began with a stronger claim—that a diffusion reconstructor proves “fabrication is a property of the objective,” a certified hallucination. A pre-submission re-analysis retracted it. We document the retraction in full because the negative result is itself informative and because selective reporting of the earlier (favourable) numbers would be exactly the failure the certificate is meant to prevent.

Setup. An arbitrary-mask conditional DDPM (12-channel binary mask + observed values; random-mask training so one model serves any configuration, with explicit leakage tests) reconstructs {V2,V4,V6} from {I,II,V1,V3,V5} under classifier-free-style guidance weight $w \in \{1, 2, 4, 6\}$ (results/gpu_diffusion_leakfixed.json, realism_metrics.json, gpu_deficit_ci.json).

(1) The original effect was a configuration-leakage artifact. The earlier exhibit reused one model trained on a fixed observed set to score a *different* configuration. With leakage removed (arbitrary-mask model + leakage guards) and 5 training seeds, the “achievability gap” $\Delta\rho = \rho_{\text{achievable}} - \rho_{\text{model}}$ is monotone in w but *marginal*: mean $\Delta\rho = -0.031, -0.024, +0.024, +0.058$ for $w = 1, 2, 4, 6$, with ± 0.044 at $w=6$ ($\approx 1.3\sigma$). The previously reported “4.3 σ ” does not reproduce.

(2) Realism does not improve with guidance—the causal premise is false. The story required higher guidance to buy realism at the cost of fidelity. Instead realism is *no better* at high guidance: from $w=1$ to $w=6$ the target-lead PSD log-distance rises (V2: 1.95 → 2.35; a small dip at $w=2$ then a clear rise, so worsening but not monotone) and the amplitude-Wasserstein distance rises (V2: 0.29 → 0.67). The trend is toward worse realism on most axes (a few, e.g. V4 amplitude, are flat); there is no monotone realism/fidelity trade to exploit.

(3) What is true instead. The predictable non-dipolar residual is real but modest: oracle correlations (gpu_oracle_gate.json) are $\rho \approx 0.49\text{--}0.55$ for V2 (QRS/ST) and 0.21–0.34 for V4/V6—consistent with the calibrated Tier-II intervals of Sec. 4. So the honest statement is a *graded predictability* of the residual, calibrated per feature and lead, not a proof of fabrication. We removed the fabrication-objective claim and rebuilt the paper around the parts that withstand scrutiny.

8 Limitations

The subspace $\mathbf{M}_s$ is estimated, so η/κ inherit its estimation error (reported via record-bootstrap CIs) and its cohort dependence (Sec. 6); the ST/T third direction is cohort-specific. Identifiability is stated for the low-rank (dipolar) coordinate: a reconstructor may still recover predictable non-dipolar structure (Sec. 5), so $\eta > 0$ bounds the dipolar component, not every waveform detail. Conformal coverage is within- (S, s, ℓ) -group marginal under exchangeability, not conditional on diagnosis; rare diagnostic subgroups are under-covered (Sec. 4). Delineation uses NeuroKit2 and inherits its errors. The certificate is a pre-reconstruction screen, not a substitute for clinical validation.

9 Conclusion

Reduced-lead ECG reconstruction has a known low-rank structure per segment. Exploiting it gives a graded, per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor, plus a total ST-threshold error rate that is empirically similar across the evaluated reconstructors. It converts “trust the reconstruction” into a per-feature, per-lead, calibrated statement, and it told us honestly when one of our own earlier claims was not supported.

A Proof of Proposition 1

Write $\mathbf{M}_{s,S} = \mathbf{Sel}_S \mathbf{M}_s$ with truncated SVD $\mathbf{M}_{s,S} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ at tolerance ϱ , where $\mathbf{U}, \mathbf{V}$ have orthonormal columns, so $\mathbf{M}_{s,S}^+ = \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^\top$ and $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S} = \mathbf{V} \mathbf{V}^\top$ is the orthogonal projector onto the retained row space $\text{span}(\mathbf{V})$ (the numerical row space of $\mathbf{M}_{s,S}$ at tolerance ϱ ; the ϱ -sensitivity sweep in Sec. 5 reports where this truncation matters). Below η, κ are understood at this ϱ .

(i) Identifiability. The dipolar part of lead ℓ is the linear functional $f_\ell(\mathbf{d}) = \mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$. From S we observe $\mathbf{Sel}_S \mathbf{M}_s \mathbf{d} = \mathbf{M}_{s,S} \mathbf{d}$, i.e. the coordinates $\mathbf{V}^\top \mathbf{d}$ (the components of $\mathbf{d}$ in $\text{row}(\mathbf{M}_{s,S})$) up to noise; the orthogonal complement $(\mathbf{I} - \mathbf{P}_{\text{obs}}) \mathbf{d}$ is unobserved. Decompose $\mathbf{e}_\ell^\top \mathbf{M}_s = \mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{P}_{\text{obs}} + \mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})$. If $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2 = 0$ then $f_\ell(\mathbf{d}) = \mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{P}_{\text{obs}} \mathbf{d}$ depends on $\mathbf{d}$ only through the observed coordinates $\mathbf{P}_{\text{obs}} \mathbf{d}$, hence is determined by $\mathbf{M}_{s,S} \mathbf{d}$: lead ℓ ’s dipolar component is identifiable. Conversely if $\eta_{s,\ell}(S) > 0$ there is a unit $\boldsymbol{\delta} \in \ker \mathbf{M}_{s,S}$ (an unobserved dipole direction, so $\mathbf{M}_{s,S} \boldsymbol{\delta} = 0$) with $\mathbf{e}_\ell^\top \mathbf{M}_s \boldsymbol{\delta} \neq 0$; the two dipoles $\mathbf{d}$ and $\mathbf{d} + t \boldsymbol{\delta}$ produce identical observations for all t but change f_ℓ by $t \mathbf{e}_\ell^\top \mathbf{M}_s \boldsymbol{\delta}$, so no estimator recovers f_ℓ at any SNR. This is the estimability criterion [8] for f_ℓ .

(ii) Conditioning. The mean-centred estimate is $\hat{\mathbf{L}}_s - \boldsymbol{\mu}_s = \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{y}_S - \boldsymbol{\mu}_{s,S})$ with $\mathbf{y}_S - \boldsymbol{\mu}_{s,S} = \mathbf{M}_{s,S} \mathbf{d} + \mathbf{Sel}_S \mathbf{r}_s + \mathbf{n}$. On the identifiable part its lead- ℓ error from the observed non-dipolar residual and noise is $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{Sel}_S \mathbf{r}_s + \mathbf{n})$, and by Cauchy–Schwarz $|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{Sel}_S \mathbf{r}_s + \mathbf{n})| \leq \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2 (\|\mathbf{Sel}_S \mathbf{r}_s\|_2 + \|\mathbf{n}\|_2) = \kappa_{s,\ell}(S) (\|\mathbf{Sel}_S \mathbf{r}_s\|_2 + \|\mathbf{n}\|_2)$, using Cauchy–Schwarz and the triangle inequality. Each per-lead conditioning is a row norm of $\mathbf{A} = \mathbf{M}_s \mathbf{M}_{s,S}^+$, so $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{A}\|_2 \leq \|\mathbf{A}\|_2 = \kappa_s(S)$; the global κ_s upper-bounds every per-lead value but is not their maximum. $\square$

B Reproducibility

Code, fixed seeds, an exact environment lock (`env.lock.txt`), and a one-command pipeline (`experiments/run_all`) are released; every table number is regenerated from result JSON (`paper/emit_baseline_table.py`),

and a CI workflow runs the test suite on every push. The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, 3 deterministic seeds, Adam 2×10^{-4} , MSE on masked leads, random-mask training); the residual predictor is a gradient-boosted quantile regressor (pinball loss) with strict fold discipline (folds 1–7 train, 9 calibrate, 10 test).

References

- [1] T. Gradowski and T. Buchner, “Deep learning model for ECG reconstruction reveals the information content of ECG leads,” *arXiv preprint arXiv:2502.00559*, 2025.
- [2] G. V. Cardoso, L. Bedin, J. Duchateau, R. Dubois, and E. Moulines, “Bayesian ECG reconstruction using denoising diffusion generative models,” *arXiv preprint arXiv:2401.05388*, 2024.
- [3] L. Edenbrandt and O. Pahlm, “Vectorcardiogram synthesized from a 12-lead ECG: superiority of the inverse Dower matrix,” *Journal of Electrocardiology*, vol. 21, no. 4, pp. 361–367, 1988.
- [4] J. A. Kors, G. van Herpen, A. C. Sittig, and J. H. van Bommel, “Reconstruction of the Frank vectorcardiogram from standard electrocardiographic leads: diagnostic comparison of different methods,” *European Heart Journal*, vol. 11, no. 12, pp. 1083–1092, 1990.
- [5] S. P. Nelwan, J. A. Kors, S. H. Meij, J. H. van Bommel, and M. L. Simoons, “Reconstruction of the 12-lead electrocardiogram from reduced lead sets,” *Journal of Electrocardiology*, vol. 37, no. 1, pp. 11–18, 2004.
- [6] A. Lence *et al.*, “ECGrecover: A deep learning approach for electrocardiogram signal completion,” in *Proc. 31st ACM SIGKDD Conf. Knowledge Discovery and Data Mining (KDD)*, 2025, arXiv:2406.16901.
- [7] X. Fang, H. Wang, J. Cai, others, and S. Hong, “ImputeECG: Deep learning reconstruction of complete 12-lead electrocardiograms from incomplete recordings for cardiac assessment,” *arXiv preprint arXiv:2607.05009*, 2026.
- [8] H. Scheffé, *The Analysis of Variance*. New York: Wiley, 1959.
- [9] S. Joshi and S. Boyd, “Sensor selection via convex optimization,” *IEEE Transactions on Signal Processing*, vol. 57, no. 2, pp. 451–462, 2009.
- [10] Y. Romano, E. Patterson, and E. J. Candès, “Conformalized quantile regression,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [11] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [12] P. Wagner, N. Strodthoff, R.-D. Bousseljot, D. Kreiseler, F. I. Lunze, W. Samek, and T. Schaeffter, “PTB-XL, a large publicly available electrocardiography dataset,” *Scientific Data*, vol. 7, no. 1, p. 154, 2020.
- [13] D. Makowski, T. Pham, Z. J. Lau *et al.*, “NeuroKit2: A Python toolbox for neurophysiological signal processing,” *Behavior Research Methods*, vol. 53, no. 4, pp. 1689–1696, 2021.

- [14] J. Zheng, J. Zhang, S. Danioko, H. Yao, H. Guo, and C. Rakovski, “A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients,” *Scientific Data*, vol. 7, no. 1, p. 48, 2020.